

Chapter 8: Attitude Maneuvers

Dr. Alexander Wittig

- Any questions on previous weeks content?

- **Introduction**
- **Chapter 1: In 20.000 km, turn right. Orbital Manoeuvres**
 - Coplanar Manoeuvres
 - Non-Coplanar Transfers
- **Chapter 2: That's mildly disturbing Orbital Perturbations & Propagation**
 - Non-Uniform Gravity Field
 - Third-Body Perturbation
 - Atmospheric Drag
 - Solar Radiation Pressure
 - Orbit Propagation Methods
- **Chapter 3: Is it a plane? Is it a bird? Orbit Observation & Determination**
 - Observation Techniques
 - Preliminary Orbit Determination
- **Chapter 4: Waltzing Matilda Relative Orbital Motion**
 - Relative Orbital Motion
 - Linearization & State Transition Matrix
- **Chapter 5: To boldly go... Interplanetary Trajectories & Optimization**
 - Lambert's Problem
 - Patched Conics
 - Gravity-assisted fly-by trajectories
- **Chapter 6: Torn between two bodies Restricted Three-Body Problem**
 - N-Body Problem
 - Circular Restricted Three-Body Problem
 - Natural Dynamics around Libration Points
- **Chapter 7: You spin me right round Attitude Dynamics**
 - Attitude representation
 - Attitude Dynamics
- **Chapter 8: In the spin cycle Attitude Motion**
 - Torque-Free Motion
 - Attitude Maneuvers
- **Summary and Revision**

- Attitude maneuvers
 - Impulsive momentum change
 - Environmental Torques
 - Coning
 - Yoyo De-spin

- Actuators & Sensors
 - Reaction Wheels, Control Moment Gyroscopes, Attitude Thrusters
 - Relative & Absolute Attitude Sensors
 - Gyroscopes, IMUs

Short time torques: approximately instantaneous

$$\frac{\Delta \vec{H}}{\Delta t} \approx \frac{d}{dt} \vec{H} = \vec{T} \Rightarrow \Delta \vec{H} \approx \vec{T} \Delta t$$

in inertial frame!

- If time scale is much shorter than rotation period
- Like impulsive delta-V maneuvers
- Possible sources

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in inertial frame!

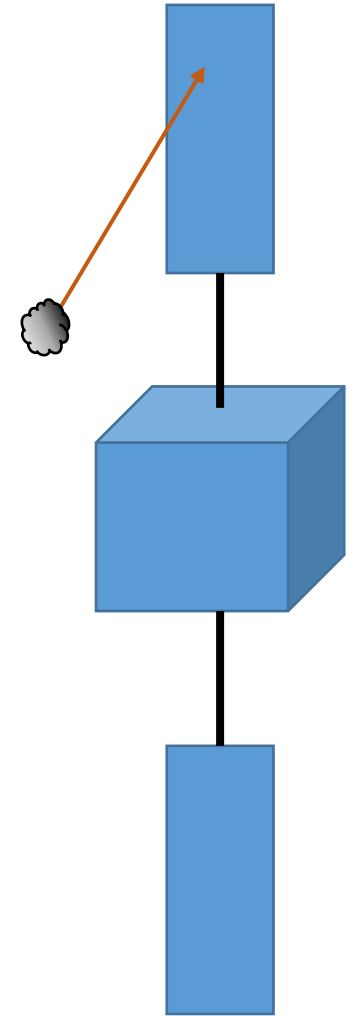
- If time scale is much shorter than rotation period
- Like impulsive delta-V maneuvers
- Possible sources
 - Micro-meteorites
 - Attitude control thrusters
 - Reaction wheels, control moment gyros

Environmental Torques

- Fast impacts
- Low mass
- Impact on both linear and rotational state

Modelling:

- Impulsive momentum change
- Embedded / blow through
- Fragmentation / no fragmentation



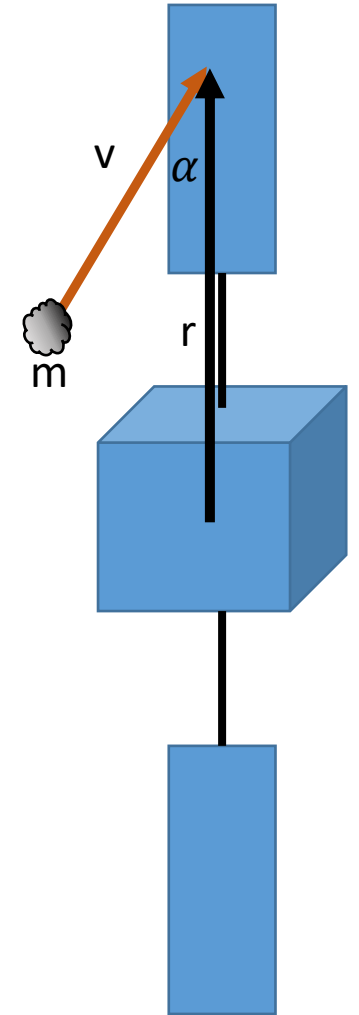
- Fast impacts
- Low mass
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Modelling:

- Impulsive momentum change
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$$\begin{aligned}\vec{F} &= m\vec{a} \approx m \frac{\Delta\vec{v}}{\Delta t} \\ \vec{T} &= \vec{r} \times \vec{F} \\ \Rightarrow \Delta\vec{H} &= m (\vec{r} \times \Delta\vec{v})\end{aligned}$$

Example: Micro-meteorite sticking to satellite, no fragmentation



- Fast impacts
- Low mass
- Impact on both linear and rotational state

Modelling:

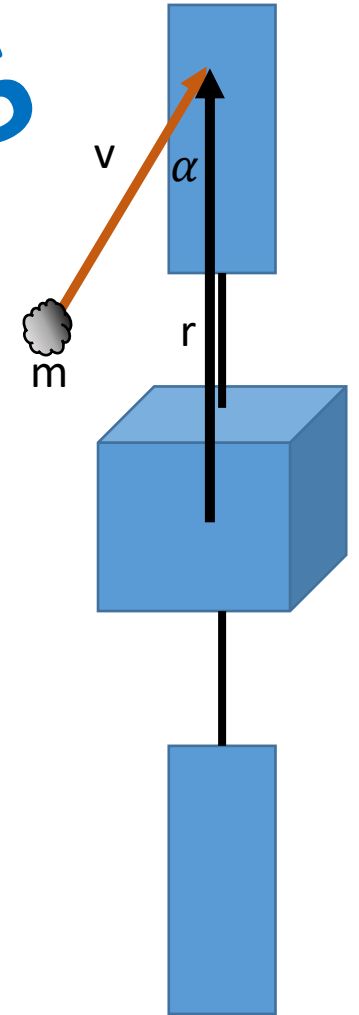
- Impulsive momentum change
- Embedded / blow off
- Fragmentation

$$\vec{F} = m\vec{a} \approx m \frac{\Delta\vec{v}}{\Delta t}$$

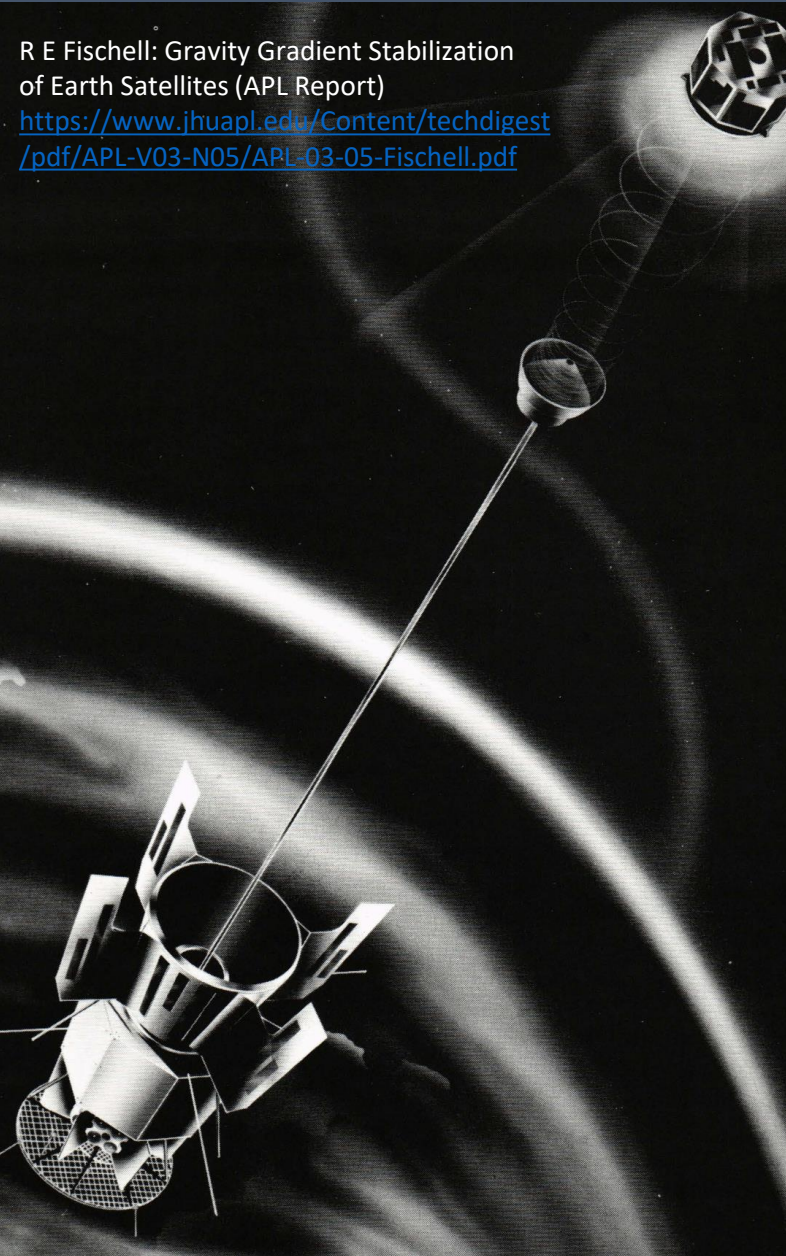
$$\vec{T} = \vec{r} \times \vec{F}$$

$$\Rightarrow \Delta\vec{H} = m (\vec{r} \times \Delta\vec{v})$$

Example: Micro-meteorite sticking to satellite, no fragmentation



see Chapter 7 Video Lecture 35



R E Fischell: Gravity Gradient Stabilization of Earth Satellites (APL Report)
<https://www.jhuapl.edu/Content/techdigest/pdf/APL-V03-N05/APL-03-05-Fischell.pdf>

$$\vec{T}_{GG} = \frac{3\mu}{r^5} (\vec{r} \times \mathbf{I}\vec{r})$$

- Gravitational field not uniform: less torque on higher part of S/C than lower
 - Principal axes aligned with gravity field are invariant
 - Smallest principal moment axis is stable
 - Larger difference between I_{123} → larger torque
- Gravity Gradient Stabilization
- Order of magnitude (LEO):
 - Cube-Sat: $< 10^{-8}$ Nm
 - Medium Satellite: $10^{-4} - 10^{-6}$ Nm
 - Space Shuttle: ~ 1 Nm
 - Large Tethered Spacecraft: > 10 Nm

Gravity Gradient



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$$\vec{T}_{GG} = \frac{3\mu}{r^5} (\vec{r} \times \mathbf{I} \vec{r})$$

- Gravitational field not uniform → torque on higher part of S/C than lower
- Principal axes of inertia & gravity field are invariant
- Smaller moment of inertia about principal axis is stable
- Difference between I_{123} → larger torque

Gravity Gradient Stabilization

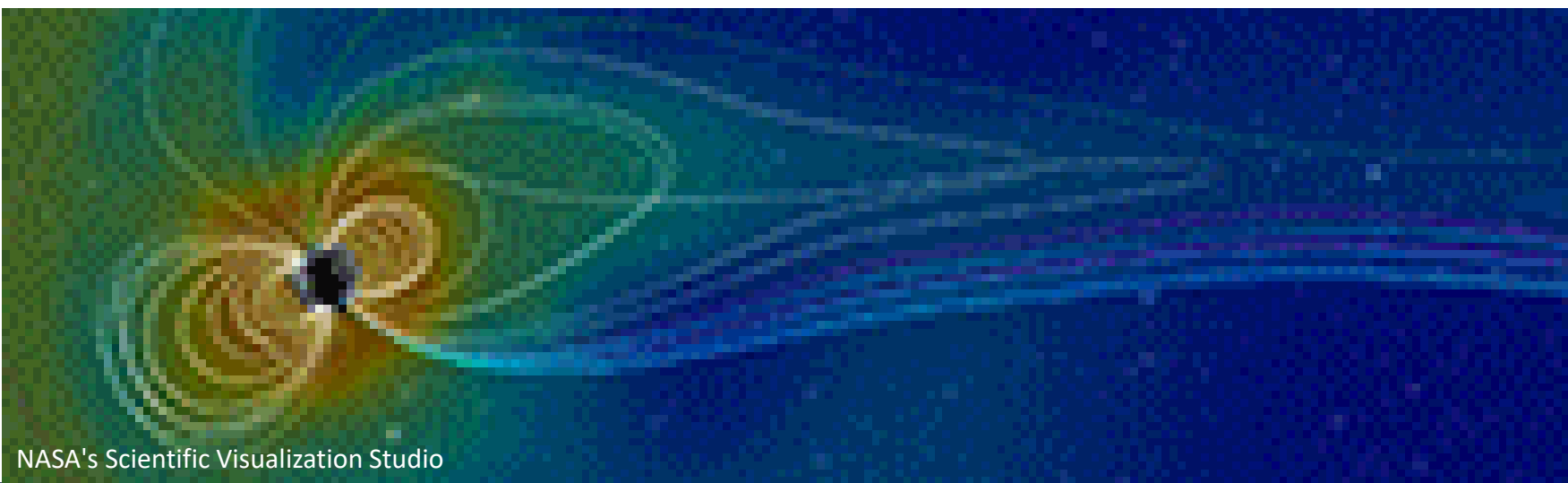
Order of magnitude (LEO):

- Cube-Sat: $< 10^{-8}$ Nm
- Medium Satellite: $10^{-4} - 10^{-6}$ Nm
- Space Shuttle: ~ 1 Nm
- Large Tethered Spacecraft: > 10 Nm

see Chapter 7 Video Lecture 36

$$\vec{T} = \vec{m} \times \vec{B}$$

- Torque from magnetic dipoles in magnetic fields
- Earth magnetic field in LEO: $\sim 10^{-5}$ Tesla = 0.1 Gauss
- Typical torque for strong penny-size Neodymium magnet: 10^{-5} Nm
- Passive attitude control (permanent or electric)



NASA's Scientific Visualization Studio



Weldon Pocket Compass and Sundial (Pitt Rivers Museum)

Magnetic Torques

$$\vec{T} = \vec{m} \times \vec{B}$$

- Torque from magnetic dipoles in magnetic fields
- Earth magnetic field in LEO: $\sim 10^{-5}$ Tesla = 0.1 Gauss
- Typical torque for strong penny-size Neodymium magnet: 10^{-5} Nm
- Passive attitude control (permanent magnet / electric)

see Chapter 7 Video Lecture 37



Other sources of torque:

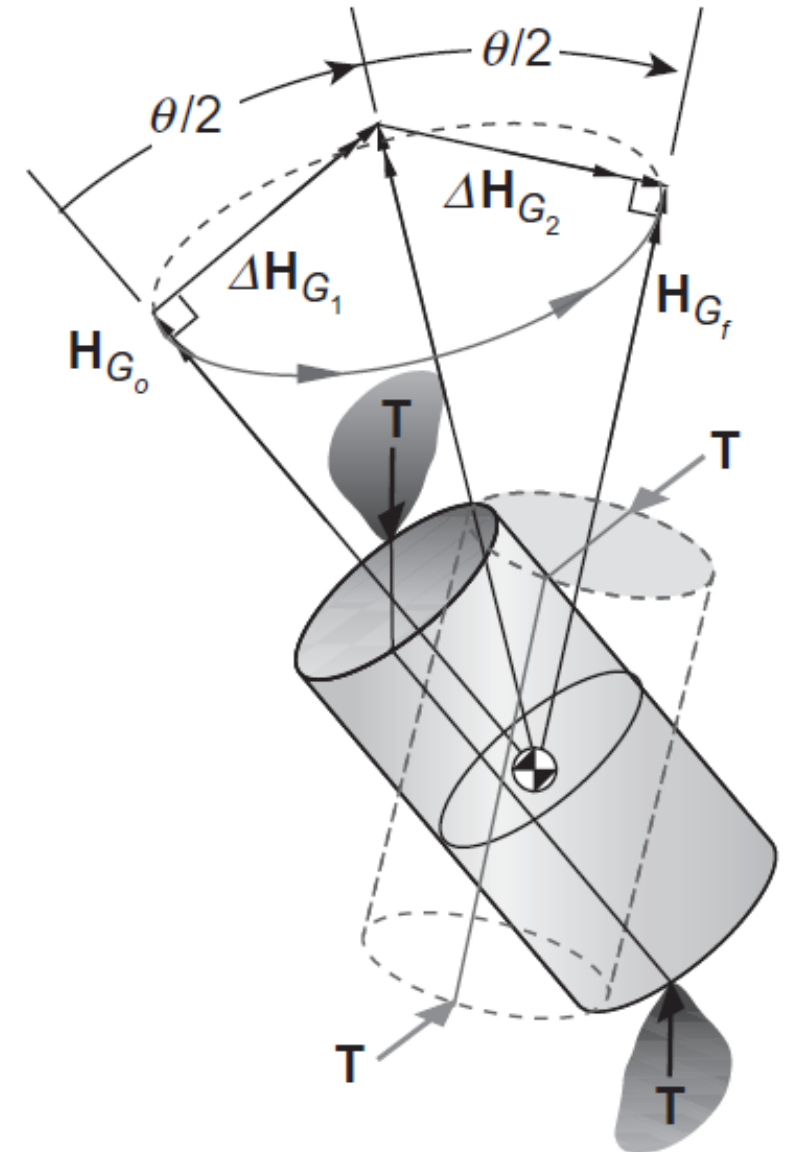
- Solar radiation torque
- Aerodynamic torque
- Electrostatic torque

Popular Science



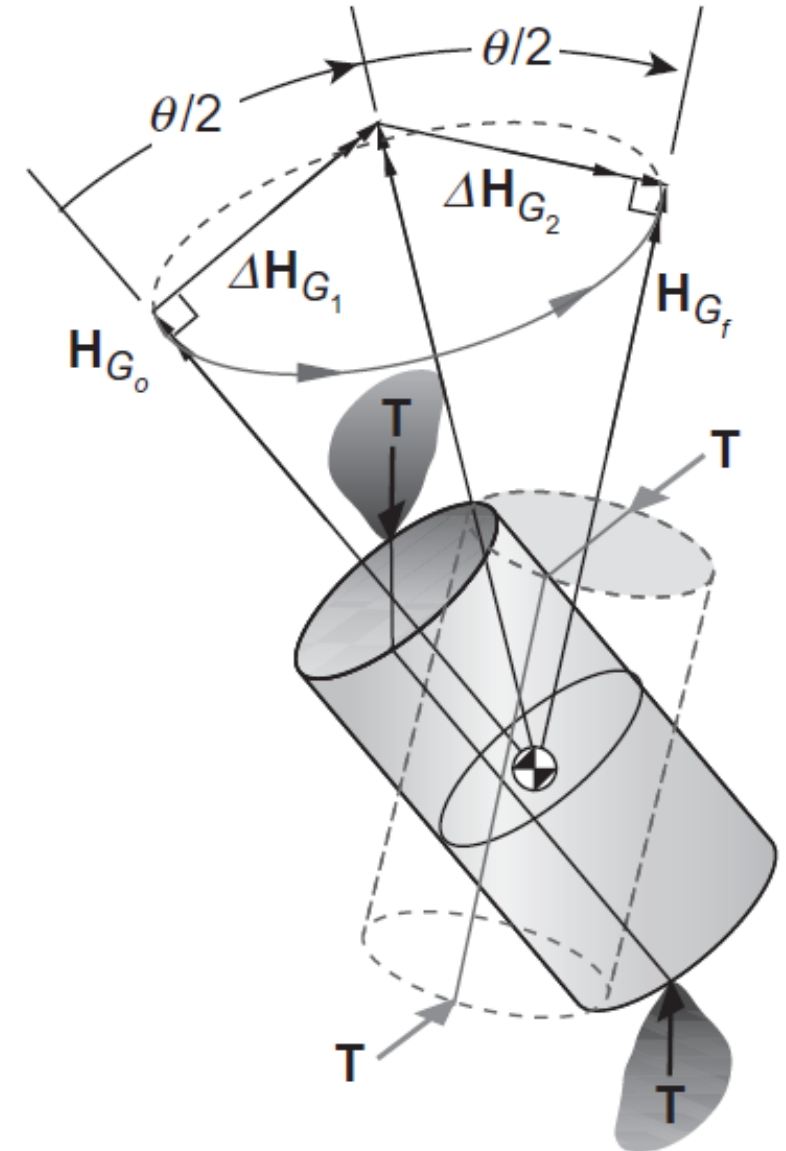
Attitude Maneuvers

Given a single-spin spin-stabilized satellite in a pure spin state (angular velocity $\omega_z = \omega_0$), how can we reorient the satellite by angle θ ?



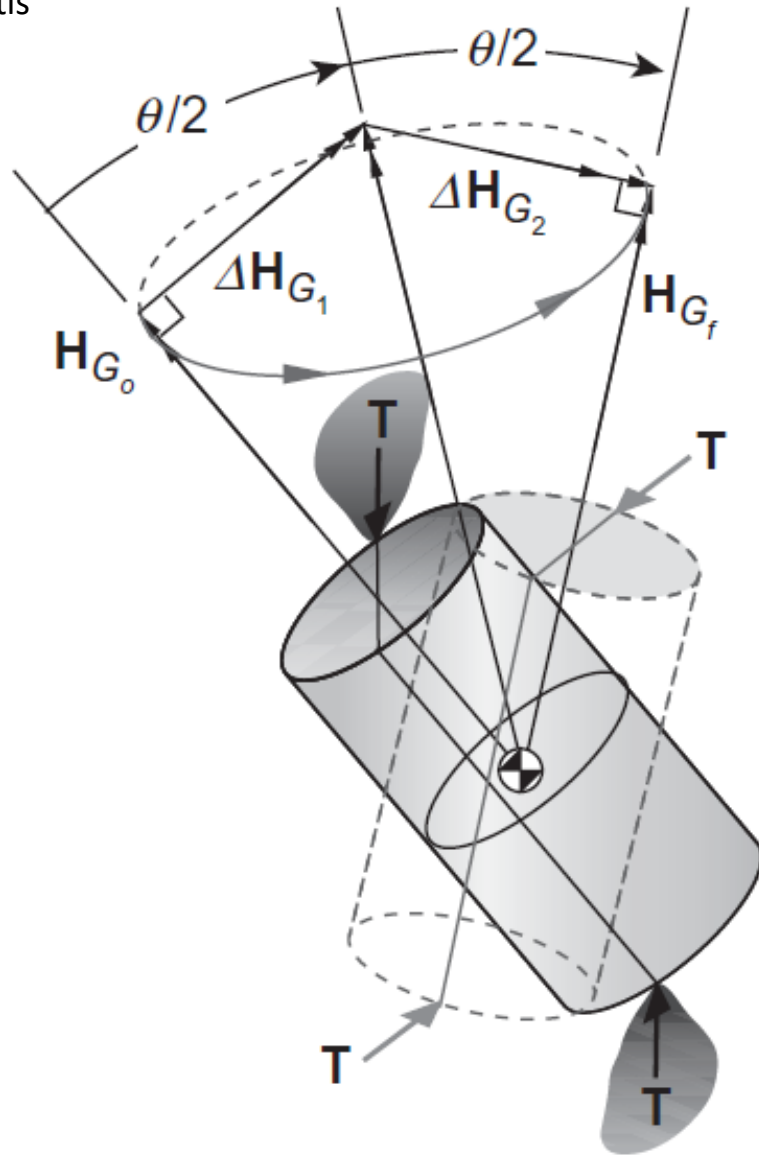
Given a single-spin spin-stabilized satellite in a pure spin state (angular velocity $\omega_z = \omega_0$), how can we reorient the satellite by angle θ ?

1. Initially, $\omega_s = \omega_0$ and $\omega_p = 0$
2. Apply $\Delta \mathbf{H}_1$ normal to $\mathbf{H}$
→ $\omega_z = \omega_0$ stays but now we have ω_s and ω_p
3. After 180° precession apply another $\Delta \mathbf{H}_2$ to realign $\mathbf{H}$ with $\mathbf{z}$



Cost of Coning Maneuver

Curtis



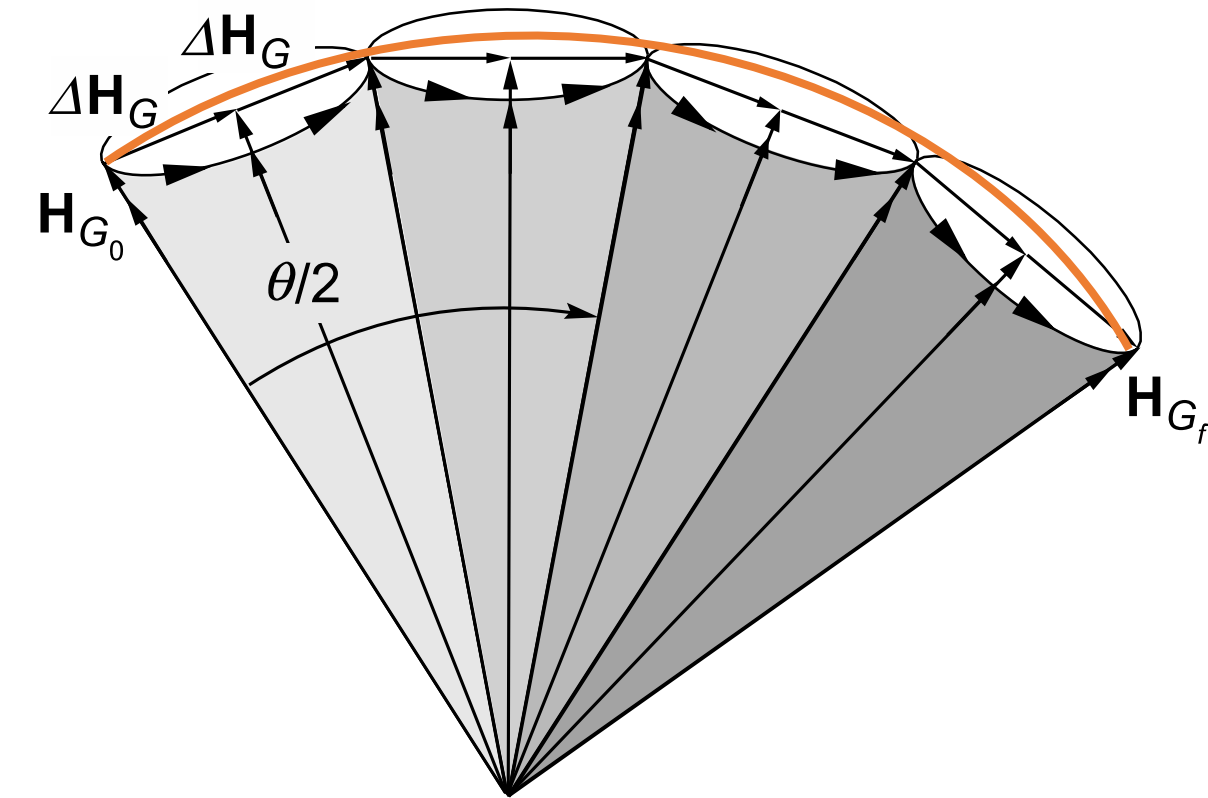
Analogous to ΔV maneuvers:
Cost is measured as total ΔH

- For single coning maneuver (from geometry):

$$\Delta H_{total} = 2 \left| \vec{H}_0 \right| \tan \frac{\theta}{2}$$

- Note: $\theta = 180^\circ$ is not possible (infinite ΔH)

Curtis



Analogous to ΔV maneuvers:
Cost is measured as total ΔH

- Repeated coning maneuvers reduce cost:

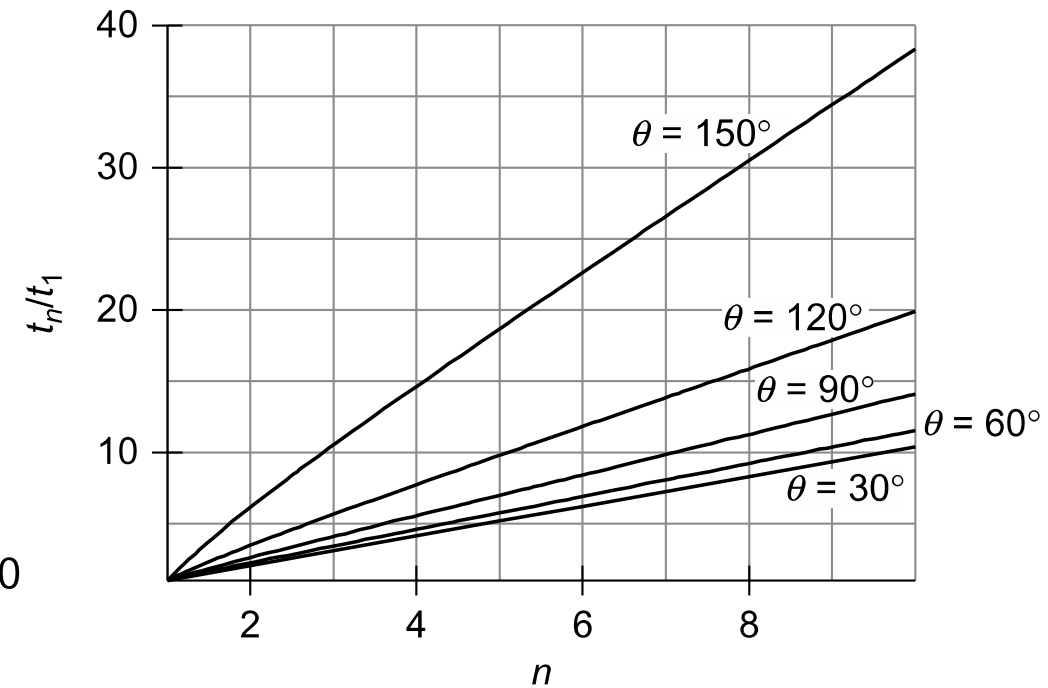
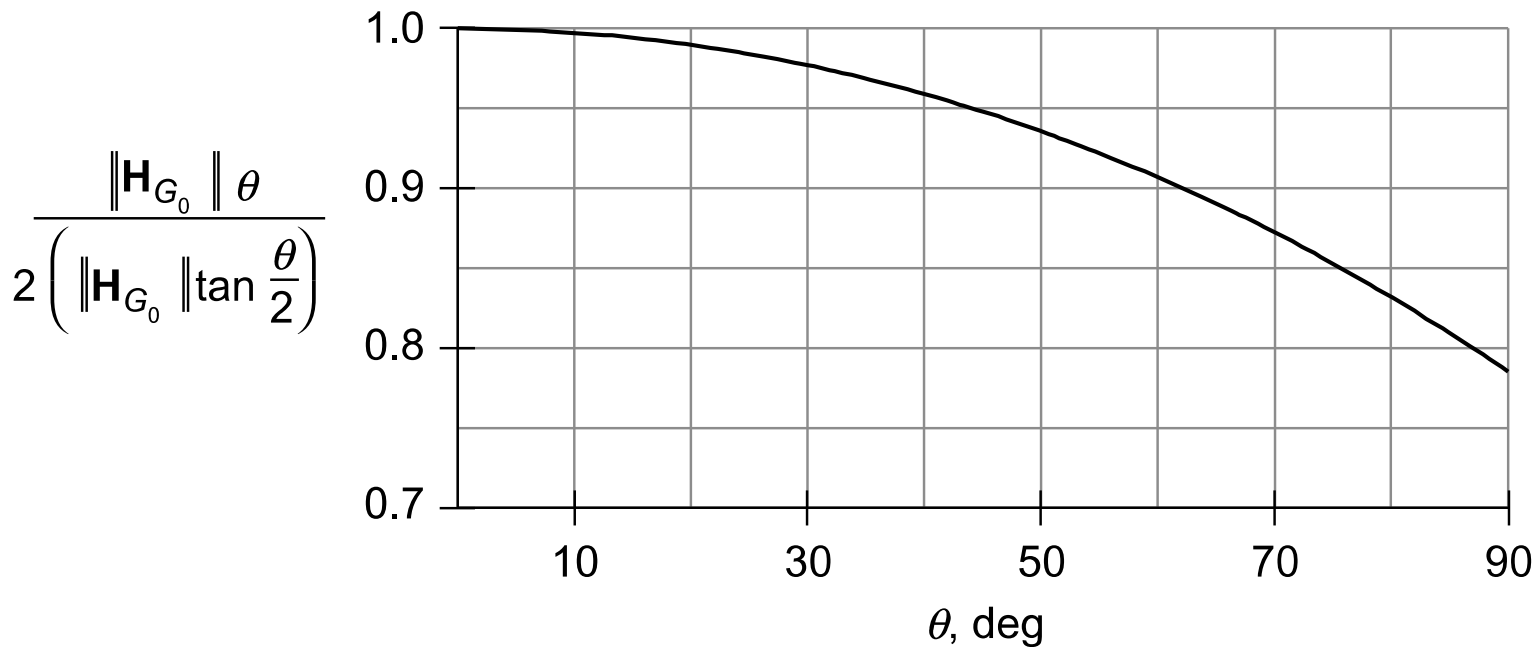
$$\Delta H_{total} \approx \left| \vec{H}_0 \right| \theta$$

(approximates arc)

- Downsides
 - time
 - maneuver accuracy

Cost of Coning Maneuver

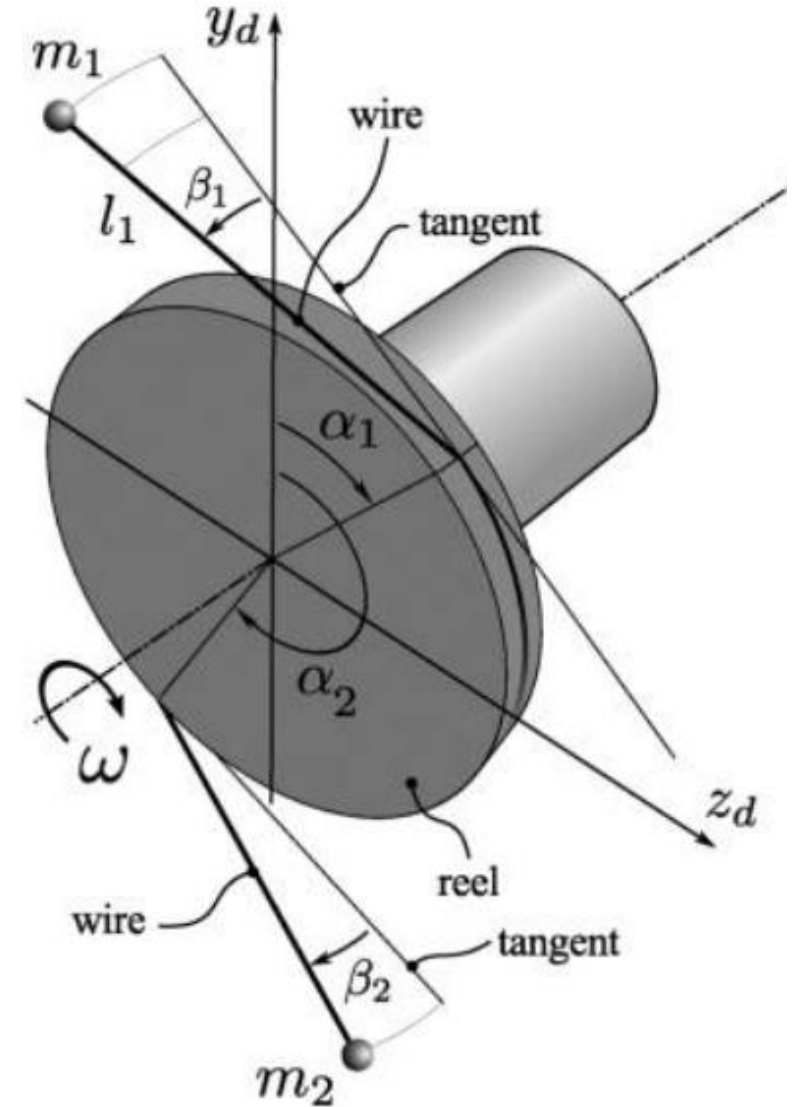
- Downsides
 - time
 - maneuver accuracy



So far: rigid bodies → fixed matrix of inertia

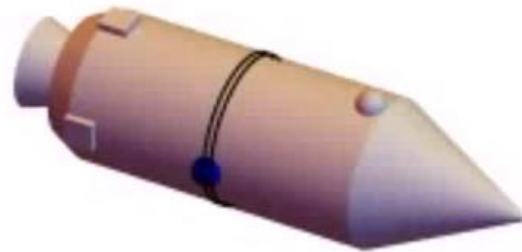
Idea: Controlled release of mass to dump angular momentum

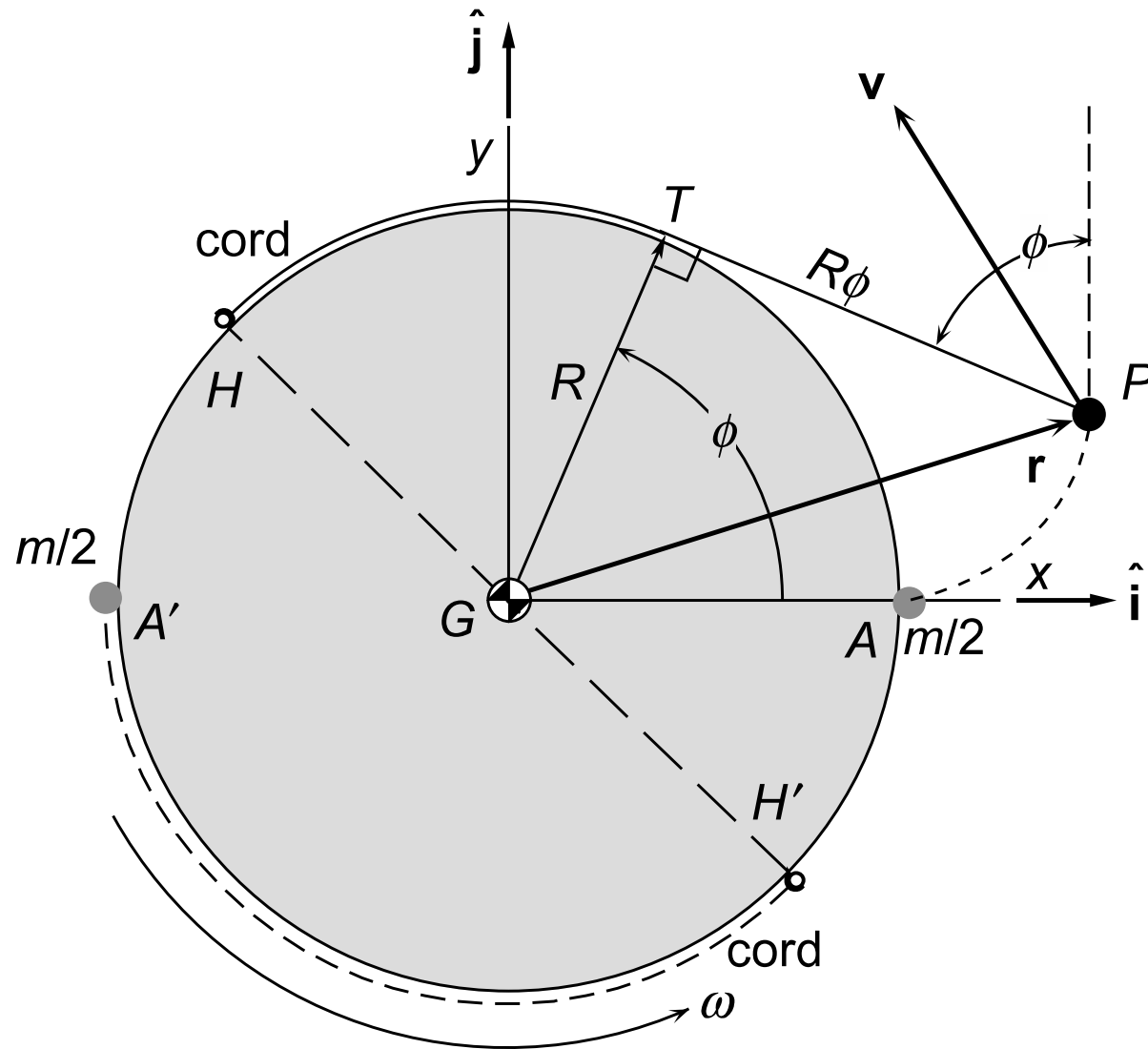
- two symmetric masses on wires
 - initially wound up
 - released at fixed length
-
- Advantages
 - fast de-spin
 - small mass requirement
 - Example: Dawn Mission
 - 3kg weights, 1.4 t S/C
 - 12 m cable
 - 46 RPM → -3 RPM



Detumbling Space Debris Using Modified Yo-Yo Mechanism
Vadim Yuditsev and Vladimir Aslanov
Journal of Guidance Control and Dynamics

0.0 sec
28.6 °/s





Goal: Determine cable length l to reduce angular velocity from ω_0 to ω .

$$K = 1 + \frac{I_3}{mR^2}$$

$$l = R \sqrt{K \frac{\omega_0 - \omega}{\omega_0 + \omega}}$$

Full de-spin:

$$l = R\sqrt{K}$$

independent of ω_0 !

Actuation

- Thrusters mounted on spacecraft
- Small thrusters: up to 500 N (Gemini spacecraft)
 - hydrogen peroxide
 - hydrazine
 - electric propulsion (especially in GEO)
- Many thrusters (6+)
(unlike translational thrusters)

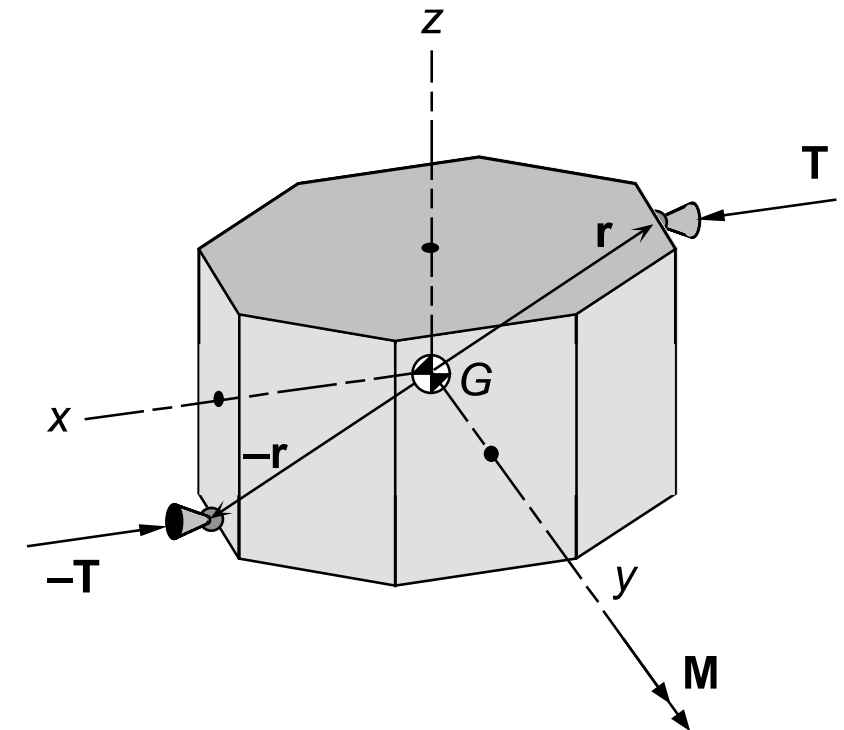


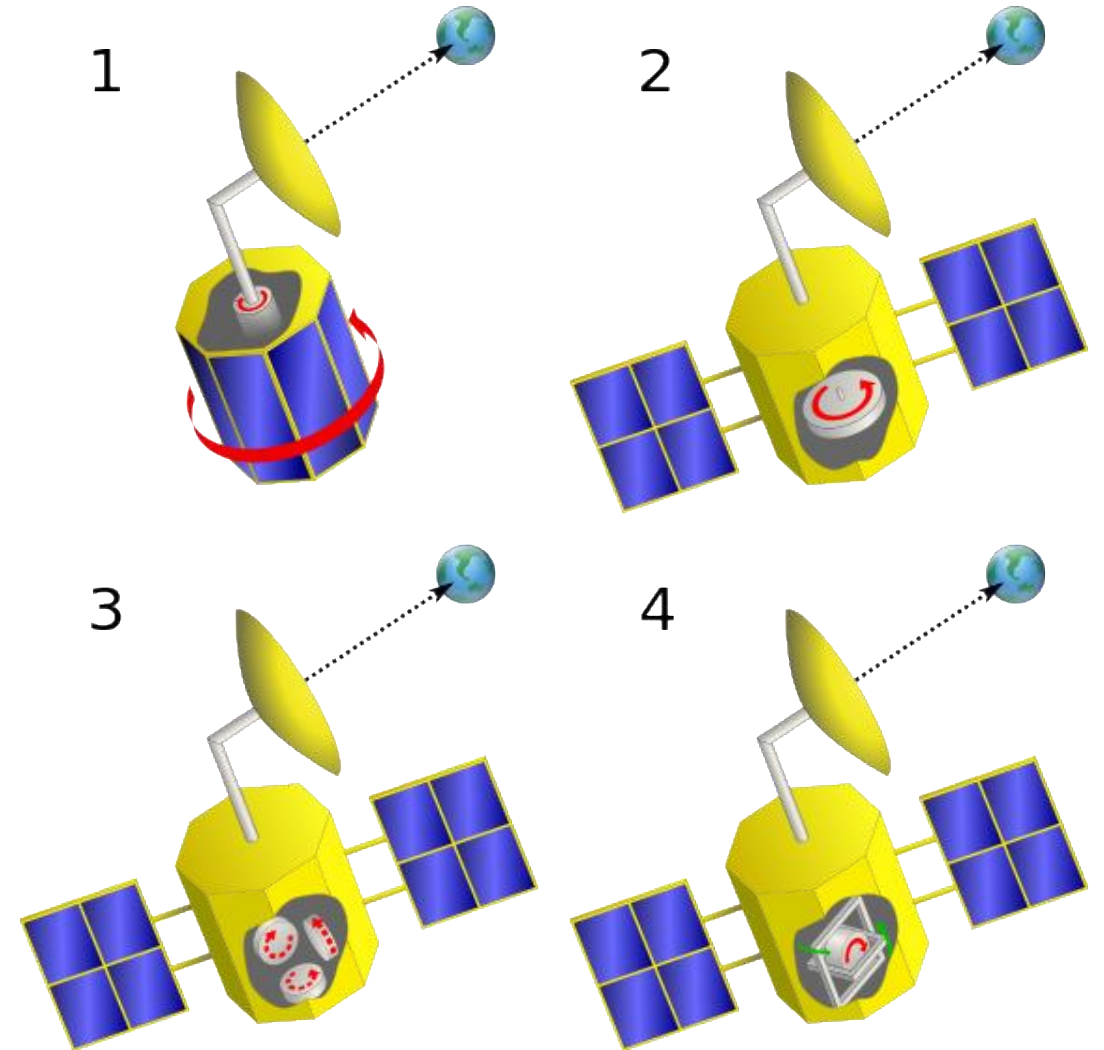
Space Shuttle (NASA)

Typical satellite layout:

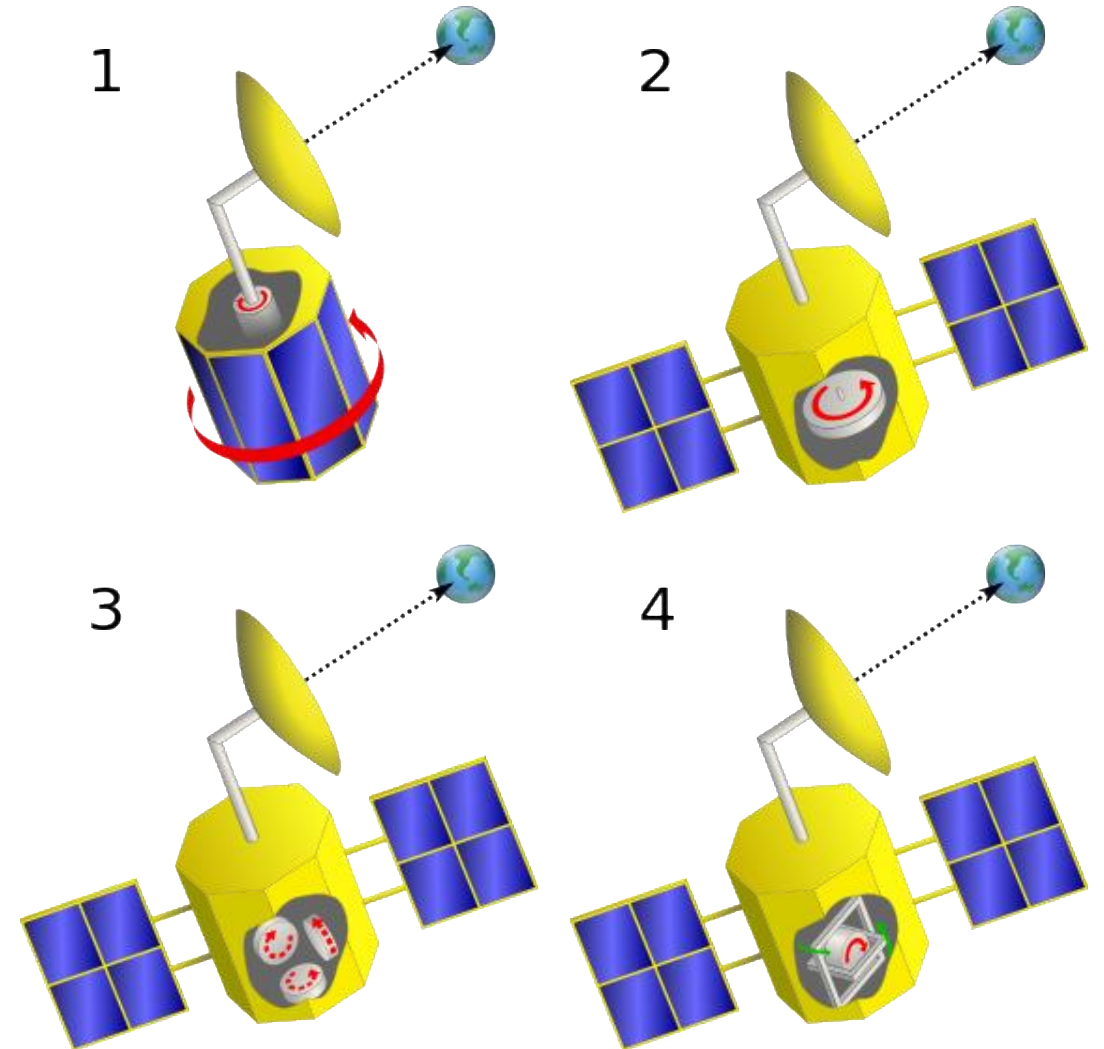
- in principal planes
one along each axis
- in pairs
to avoid linear displacement
- symmetric relative to center of mass

Typically modelled as impulsive ΔH

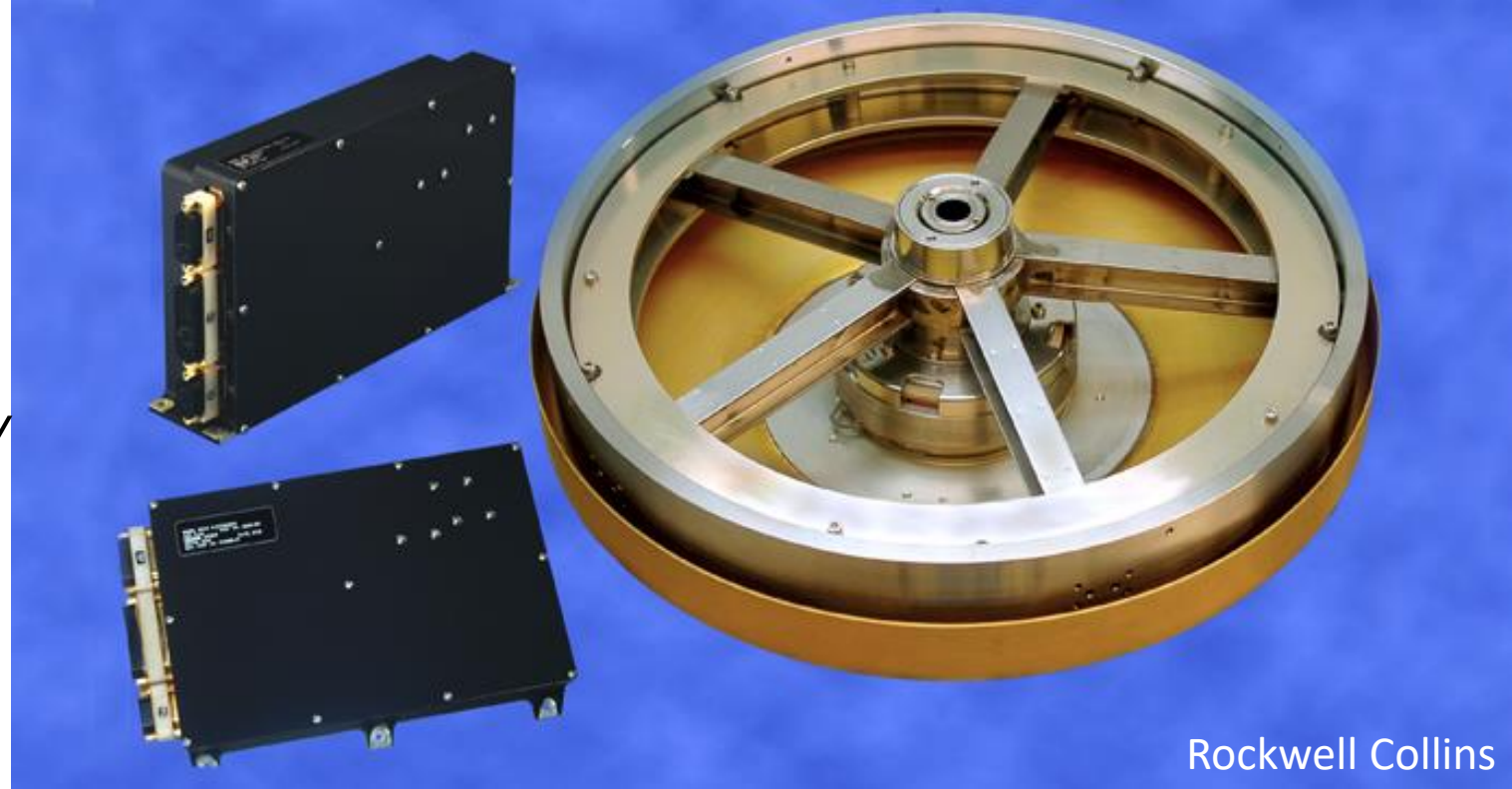
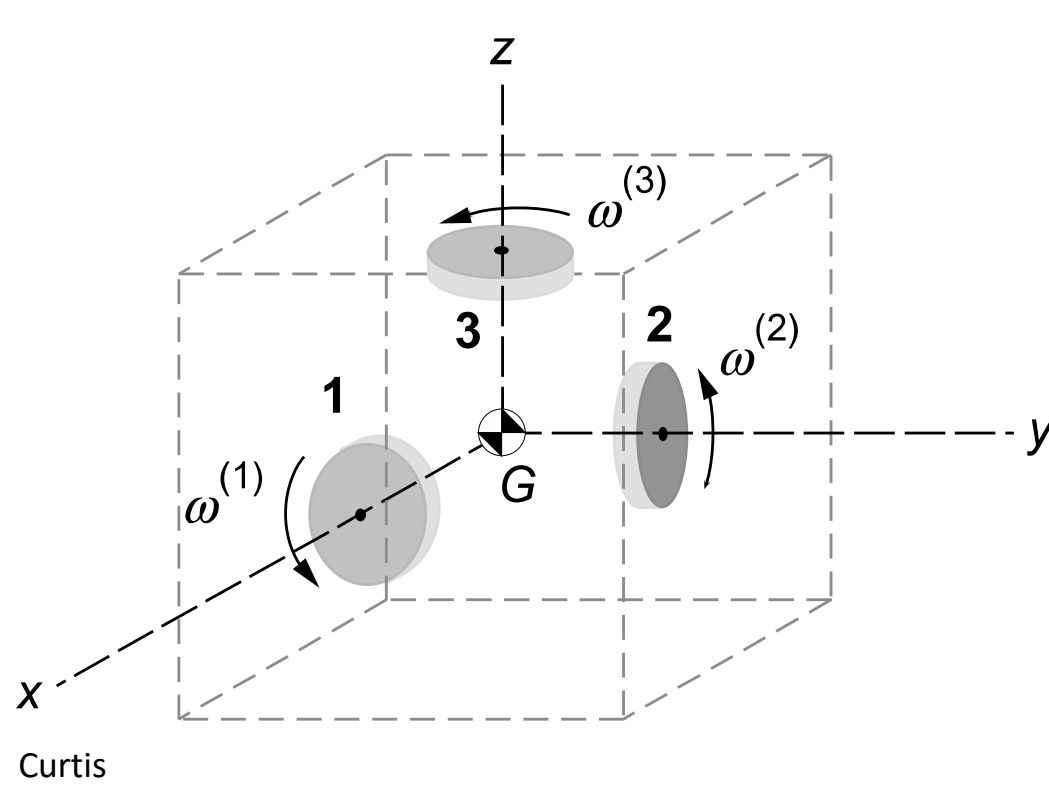




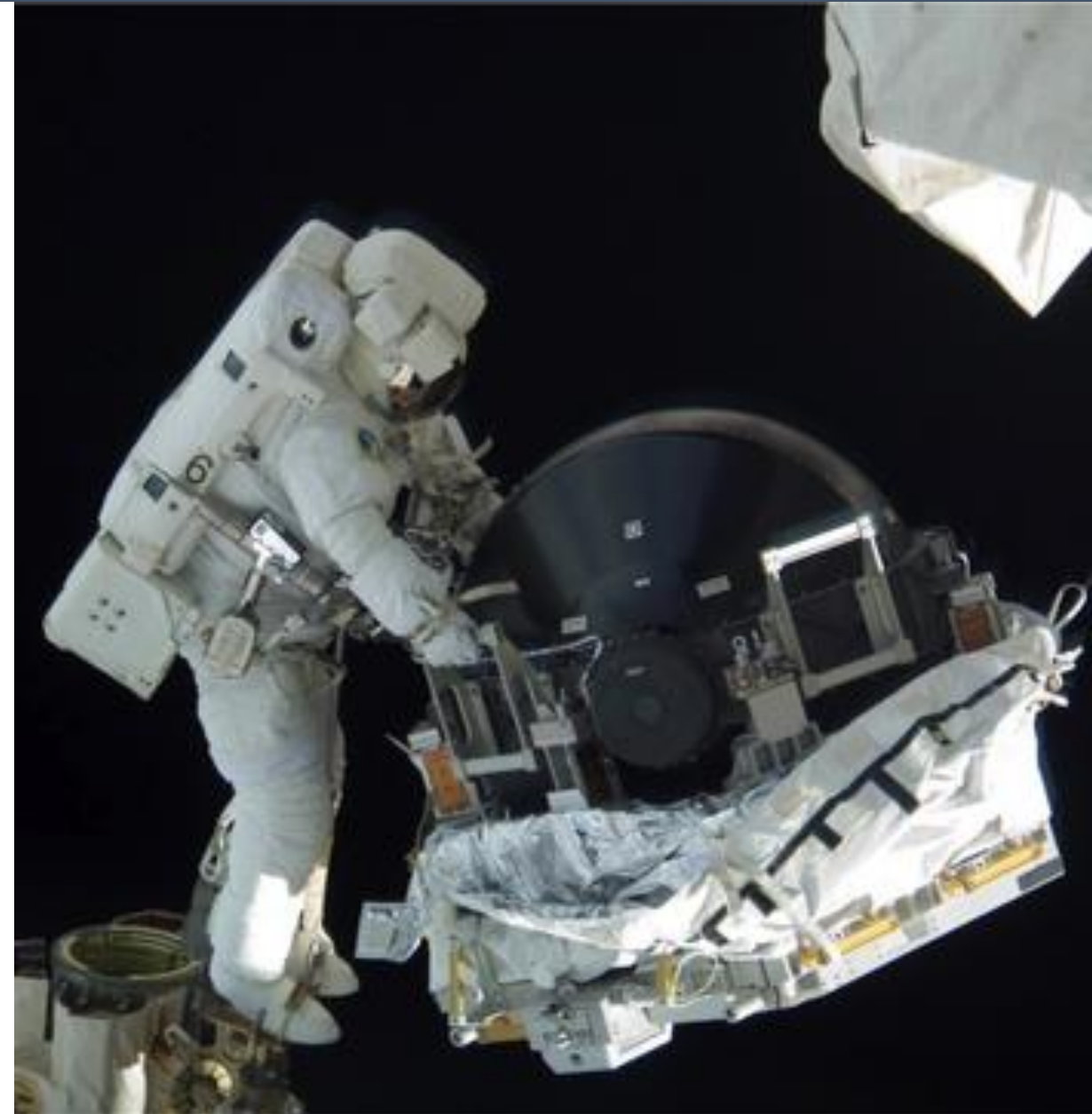
- 1. Dual spin satellite**
type of spin stabilization, body spinning, platform de-spun
- 2. Reaction Wheel**
single reaction wheel for single axis control
- 3. Reaction Wheel Assembly**
3 (or more) reaction wheels for 3-axis control
- 4. Control Moment Gyro**
1 (or more) control moment gyros for 3-axis control



- action = reaction
- total angular momentum conserved
- spin up of wheel $\rightarrow$ spin down of satellite around given axis



- Similar to Reaction Wheel
- Adjustable spin axis on gimbal
- Typically spun up to fixed, high speed (6000+ rpm)
- up to 1000+ Nm torque
- Moment exchange device
 - requires regular de-saturation
 - regular thruster burns
 - magnetic de-torque



Sensing

Relative sensors

- measure **change** in satellite attitude over time
- require initial attitude
- cheap, robust, simple
- drift over time

Absolute sensors

- detect satellite attitude with respect to **external objects**
- can initialize from zero
- often only partial attitude
- more complex

■ Gyroscopes:

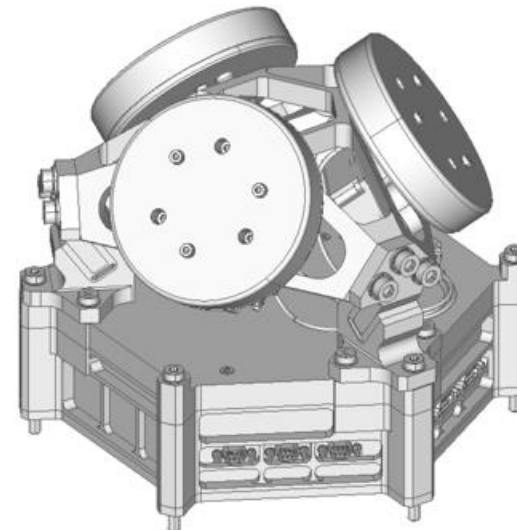
- spinning mass (classical)
- ring laser gyro (speed of light)
- MEMS (microelectromechanical system, vibrating mass)

■ IMU: Inertial Measurement Unit Complete device including

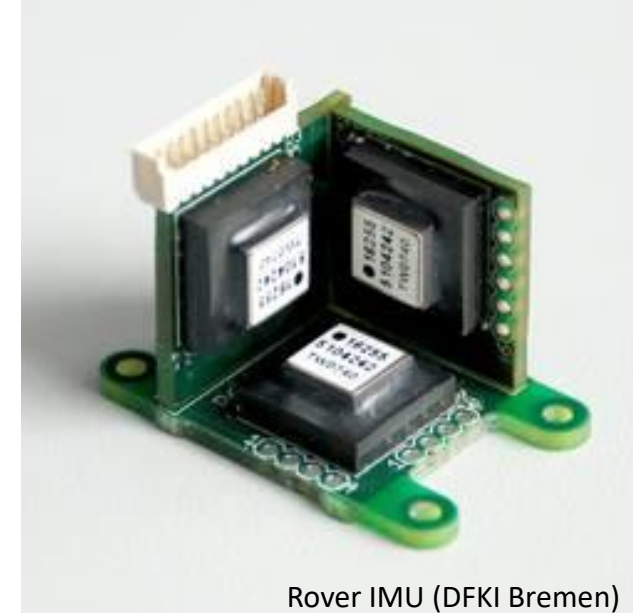
- MEMS gyros (single/multi axis)
- Filter
- Integrator
- Output: satellite attitude (e.g. roll, pitch, yaw)



3 axis ring laser gyro
(Honeywell)

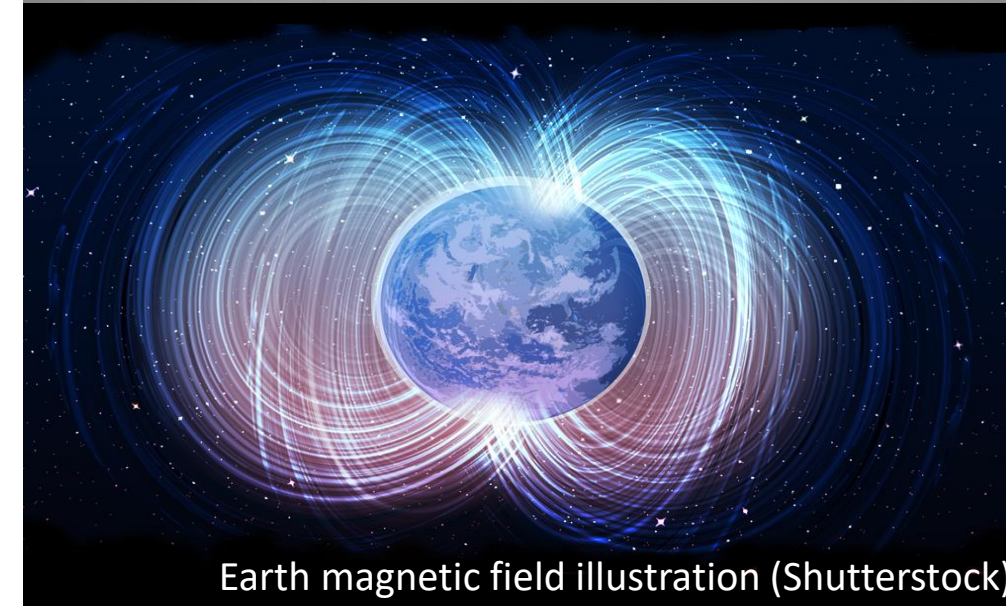


S/C IMU breadboard (ESA)



Rover IMU (DFKI Bremen)

- **Star trackers**
camera image of sky matched to pre-programmed database
- **Magnetometers**
measurement of field lines matched with position and field model (like compass)
- **Horizon sensors**
camera image to detect location of Earth horizon
- **Sun sensors**
light sensor to find bright spot in the sky



Earth magnetic field illustration (Shutterstock)

- Impulsive momentum change
- Coning maneuver
- Maneuver cost
- Yoyo de-spin
- Stabilization methods
- RCS
- CMG
- Relative/Absolute attitude sensing
- Gyroscopes/IMUs
- Star trackers, horizon sensors, sun sensors

Chapter 8: Attitude Maneuvers

Dr. Alexander Wittig